TASK-5 : CONDITIONAL LOGIC (CASE WHEN)

Q1) Categorize projects based on project duration.

Query:

```
SELECT project_id,
    project_name,
    start_date,
    deadline,
    CASE
        WHEN (deadline - start_date) < 180 THEN 'Short-Term'
        ELSE 'Long-Term'
    END AS Duration_Category
FROM project;
```

Explanation:
This query categorizes projects based on their duration. It calculates the difference between the deadline and start date and labels projects as “Short-Term” if the duration is less than 180 days; otherwise, they are labeled as “Long-Term”.

Enter a SQL expression to filter results (use Ctrl+Space)

	project_id	project_name	start_date	deadline	duration_category
1	301	Website Redesign	2024-01-05	2024-06-30	Short-Term
2	302	Mobile Application	2024-02-01	2024-08-15	Long-Term
3	303	ERP Deployment	2024-01-20	2024-12-10	Long-Term
4	304	Cloud Migration	2024-03-10	2024-09-25	Long-Term
5	305	Data Warehouse	2024-02-15	2024-11-30	Long-Term
6	306	Cybersecurity Audit	2024-04-01	2024-07-31	Short-Term
7	307	CRM Upgrade	2024-03-05	2024-10-15	Long-Term
8	308	AI Chatbot	2024-05-01	2024-12-01	Long-Term
9	309	E-commerce Portal	2024-01-25	2024-09-10	Long-Term
10	310	Payment Integration	2024-02-10	2024-07-20	Short-Term
11	311	HR Automation	2024-03-15	2024-08-30	Short-Term
12	312	Inventory System	2024-01-30	2024-09-05	Long-Term
13	313	Analytics Dashboard	2024-02-20	2024-10-30	Long-Term
14	314	Networking Upgrade	2024-04-10	2024-11-15	Long-Term
15	315	IoT Monitoring	2024-05-05	2024-12-20	Long-Term
16	316	Digital Marketing	2024-01-10	2024-06-15	Short-Term
17	317	Payroll System	2024-02-25	2024-09-30	Long-Term
18	318	Document Management	2024-03-01	2024-10-10	Long-Term
19	319	Video Streaming App	2024-04-15	2024-12-05	Long-Term

Q2) Assign a target label to departments based on department type.

Query:

```
SELECT dept_id,
    dept_name,
    CASE
        WHEN dept_name LIKE '%Software%' OR dept_name LIKE '%IT%' THEN 'Technical'
        WHEN dept_name LIKE '%Finance%' OR dept_name LIKE '%Business%' THEN
'Business'
        ELSE 'Administrative'
    END AS Department_Type
FROM department;
```

Explanation:
This query assigns a department type label based on the department name. Departments containing "Software" or "IT" are categorized as Technical, those containing "Finance" or "Business" are labeled as Business, and all remaining departments are categorized as Administrative.

	dept_id	dept_name	department_type
1	101	Human Resources	Administrative
2	102	Finance	Business
3	103	Sales	Administrative
4	104	Marketing	Administrative
5	105	Operations	Administrative
6	106	Administration	Administrative
7	107	IT Support	Technical
8	108	Software Development	Technical
9	109	Quality Assurance	Administrative
10	110	DevOps	Administrative
11	111	Cyber Security	Administrative

Q3) Determine project age status based on start date.

Query:

```
SELECT project_id,
      project_name,
      start_date,
      CASE
         WHEN start_date < '2024-01-01' THEN 'Older Project'
         ELSE 'Recent Project'
      END AS Project_Status
FROM project;
```

Explanation:
This query classifies projects based on their start date. Projects that started before January 1, 2024 are labeled as "Older Project", while those starting on or after that date are labeled as "Recent Project".

project 1

SELECT project_id, project_name, start_dat | Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

	project_id	project_name	start_date	project_status
1	301	Website Redesign	2024-01-05	Recent Project
2	302	Mobile Application	2024-02-01	Recent Project
3	303	ERP Deployment	2024-01-20	Recent Project
4	304	Cloud Migration	2024-03-10	Recent Project
5	305	Data Warehouse	2024-02-15	Recent Project
6	306	Cybersecurity Audit	2024-04-01	Recent Project
7	307	CRM Upgrade	2024-03-05	Recent Project
8	308	AI Chatbot	2024-	t
9	309	E-commerce Portal	2024-01-25	Recent Project
10	310	Payment Integration	2024-02-10	Recent Project
11	311	HR Automation	2024-03-15	Recent Project
12	312	Inventory System	2024-01-30	Recent Project
13	313	Analytics Dashboard	2024-02-20	Recent Project
14	314	Networking Upgrade	2024-04-10	Recent Project

Value: 2024-04-01
Timezone: GMT+05:30